

Reducing Linear Regression Computation Time Using Quantum Mechanics Principles

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ABSTRACT

Linear regression is one of the most efficiently used type of supervised learning algorithm in machine learning which is used as a fundamental algorithm in data analysis and machine learning. It's computational cost becomes prohibitive for high-dimensional data and large datasets due to matrix inversion and iterative optimization requirements. Quantum computing introduces new computational paradigms based on quantum mechanics principles such as superposition, entanglement, and quantum parallelism. This paper explores how quantum algorithms, particularly those inspired by the Harrow–Hassidim–Lloyd (HHL) algorithm, can significantly reduce the computation time of linear regression. We analyze classical linear regression complexity, describe quantum-based approaches, and compare their theoretical performance. The results indicate that quantum linear regression has the potential to achieve exponential speedups under certain conditions, offering promising implications for large-scale machine learning applications.

KEYWORDS – quantum mechanics and computing, linear regression, HHL algorithm, computation complexity, quantum machine learning (QML)

INTRODUCTION

Linear regression is a supervised learning (the model trains on labelled data) and statistical modelling technique that describes the relationship between a dependent variable and one or more independent variables by fitting a linear equation to observed data. The model estimates parameters by minimizing the difference between the predicted and actual values, by typically using the least squares method, gradient descent which are used for prediction, inference, and analysis of variable relationships for the given problem. Linear regression is widely applied across many fields for modeling relationships between variables and making predictions. It is commonly used in business and economics etc.

Given a dataset with m samples and n features, linear regression is defined as:

$$y = X\beta + \epsilon$$

where:

- $X \in \mathbb{R}^{m \times n}$ is the feature matrix,
- $\beta \in \mathbb{R}^n$ is the parameter vector,
- $y \in \mathbb{R}^m$ is the output vector,

In the classical approach, the regression parameters are obtained by solving the normal equation:

$$\beta = (X^T X)^{-1} X^T y$$

where $X \in \mathbb{R}^{m \times n}$ is the feature matrix, $y \in \mathbb{R}^m$ is the output vector, and $\beta \in \mathbb{R}^n$ is the vector of coefficients.

However, the classical approach is computationally expensive. For high-dimensional datasets, computing $X^T X$ scales as $O(mn^2)$ and inverting the resulting matrix scales as $O(n^3)$. Iterative methods, such as gradient descent, can reduce memory usage but require many iterations to converge.

Quantum computing offers an alternative paradigm, potentially to perform calculations exponentially faster than classical computers. Quantum computing presents a fundamentally different computational framework that leverages the principles of quantum mechanics to achieve potentially exponential speedups over classical computing. By employing quantum algorithms capable of performing specific linear algebra operations in logarithmic time with respect to problem size, quantum computing emerges as a promising approach for accelerating linear regression tasks.

The least squares method is a mathematical optimization technique used in linear regression to estimate model parameters by minimizing the sum of the squared differences between observed values and predicted values.

Mathematical Formulation

For a dataset with m samples and n features, linear regression predicts y_i as:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_n x_{in} + \epsilon_i$$

where ϵ_i is the residual error. In matrix notation:

$$y = X\beta + \epsilon$$

The optimal β minimizes the mean squared error (MSE):

$$\text{MSE} = \frac{1}{m} \sum_{i=1}^m (y_i - x_i^T \beta)^2$$

Solving the normal equation gives the least-squares solution.

Computational Complexity

The steps in classical linear regression include:

1. Computing $X^T X \rightarrow O(mn^2)$
2. Inverting $X^T X \rightarrow O(n^3)$
3. Computing $X^T y \rightarrow O(mn)$

For large n and m , these operations dominate runtime. Iterative methods like gradient descent require $O(kn^2)$, where k is the number of iterations.

Thus, classical methods are increasingly impractical for very large datasets.

By leveraging quantum phenomena such as superposition and entanglement, QML aims to accelerate learning algorithms, improve optimization, and efficiently handle high-dimensional data.

Superposition

Superposition is a fundamental principle of quantum mechanics stating that a quantum system can exist in multiple states simultaneously until it is measured. Unlike a classical bit, which is either 0 or 1, a quantum bit

(qubit) can be in a linear combination of both states at the same time. Mathematically, a qubit is represented as

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

where α and β are complex probability amplitudes and $|\alpha|^2 + |\beta|^2 = 1$. This means that upon measurement, the qubit collapses to either state $|0\rangle$ or $|1\rangle$ with probabilities determined by these amplitudes. Superposition enables quantum computers to process a vast number of possible states in parallel, providing a key source of their computational power. When multiple qubits are placed in superposition, the system can represent an exponential number of states simultaneously, forming the basis for quantum speedups in algorithms such as quantum search, simulation, and machine learning.

This allows a quantum computer to evaluate many possibilities simultaneously.

Entanglement

Entanglement is a uniquely quantum mechanical phenomenon in which two or more quantum particles become correlated in such a way that the state of each particle cannot be described independently of the others, regardless of the distance separating them. Entanglement is a correlation between qubits that enables complex operations on composite quantum systems.

Quantum Parallelism

Quantum parallelism is a key feature of quantum computing that allows a quantum system to evaluate many possible inputs simultaneously by exploiting the principle of superposition. When a quantum algorithm is applied to qubits in superposed states, the computation operates on all basis states at once, effectively performing parallel calculations within a single quantum operation. This capability enables quantum algorithms to process exponentially many states in one step, forming the foundation for potential speedups over classical algorithms.

Quantum Linear Regression Approach

Quantum linear regression aims to solve the regression problem more efficiently by exploiting quantum mechanical principles and quantum linear system solvers. Instead of explicitly computing regression coefficients using matrix inversion, the problem is transformed into a linear system that can be solved using quantum algorithms such as the Harrow–Hassidim–Lloyd (HHL) algorithm.

Reformulating as a Linear System

Given a dataset with feature matrix $X \in \mathbb{R}^{m \times n}$ and output vector $y \in \mathbb{R}^m$, the classical linear regression solution is obtained by solving the normal equation:

$$\beta = (X^T X)^{-1} X^T y$$

Let:

$$A = X^T X, b = X^T y$$

Then, the regression problem reduces to solving the linear system:

$$A\beta = b$$

Here, A is a symmetric and positive semi-definite matrix. Solving this system is computationally expensive classically, especially when the number of features n is large.

Quantum Representation of Data

1) Quantum State Encoding

In quantum computing, classical vectors are represented as quantum states. The vector b is encoded as:

$$|b\rangle = \frac{1}{\|b\|} \sum_{i=1}^n b_i |i\rangle$$

Similarly, the solution vector β is represented as a quantum state:

$$|\beta\rangle = \frac{1}{\|\beta\|} \sum_{i=1}^n \beta_i |i\rangle$$

Efficient data loading is assumed through **Quantum Random Access Memory (QRAM)**, enabling state preparation in logarithmic time.

The Harrow–Hassidim–Lloyd (HHL) algorithm is a quantum algorithm designed to solve linear systems of the form:

$$Ax = b$$

The algorithm outputs a quantum state proportional to the solution $x = A^{-1}b$.

Key Assumptions:

- Matrix A is sparse.
- Matrix A is well-conditioned.
- Efficient QRAM is available.

State Preparation

The input vector b is encoded into a quantum state:

$$|b\rangle = \sum_i b_i |i\rangle$$

This step leverages QRAM to load classical data into quantum registers.

Hamiltonian Simulation

The matrix A is treated as a Hamiltonian operator. The quantum system evolves under A as:

$$e^{iAt}$$

Efficient Hamiltonian simulation is possible when A is sparse. This step is critical for eigenvalue estimation.

Quantum Phase Estimation (QPE)

Quantum Phase Estimation is applied to estimate the eigenvalues λ_i of matrix A :

$$A |u_i\rangle = \lambda_i |u_i\rangle$$

The state after QPE becomes:

$$\sum_i \alpha_i |u_i\rangle |\lambda_i\rangle$$

where $|u_i\rangle$ are eigenvectors of A .

Controlled Rotation

A controlled rotation is applied based on the inverse of the eigenvalues:

$$|\lambda_i\rangle \rightarrow \frac{1}{\lambda_i}$$

This step effectively computes the inverse of matrix A in the quantum domain.

Step 5: Uncomputation

The eigenvalue register is uncomputed to remove auxiliary states, leaving:

$$|\beta\rangle = A^{-1} |b\rangle$$

This state encodes the regression coefficients.

Output Interpretation and Prediction

Unlike classical algorithms, quantum linear regression does not return explicit coefficient values. Instead, it produces a quantum state $|\beta\rangle$.

Predictions for a new input x are computed as:

$$\hat{y} = \langle x | \beta \rangle$$

This inner product can be estimated efficiently using quantum measurement techniques such as the **swap test**.

Computational Complexity Analysis

Algorithm	Time Complexity
Classical Matrix Inversion	$O(n^3)$
Gradient Descent	$O(kn^2)$
Quantum Linear Regression (HHL)	$O(\log n)$

The exponential speedup is achievable when:

- A is sparse
- Condition number κ is small
- Data loading is efficient

ADVANTAGES OF QML

Quantum Machine Learning (QML) offers several potential advantages over classical machine learning by leveraging quantum phenomena such as superposition, entanglement, and quantum parallelism. First, QML can process and analyze high-dimensional data more efficiently, as qubits can represent exponentially many states simultaneously, enabling faster computation for complex tasks. Second, quantum algorithms can accelerate linear algebra operations, such as matrix inversion and eigenvalue decomposition, which are fundamental to many machine learning models.

1. **Exponential Speedup:** Logarithmic dependence on problem size.
2. **Efficient High-Dimensional Processing:** Suitable for large feature spaces.
3. **Parallel Computation:** Leverages quantum superposition.
4. **Prediction-Focused Output:** Ideal for inference tasks without full parameter extraction.

Limitations and Practical Considerations

- Requires fault-tolerant quantum hardware.
- QRAM is theoretical in current implementations.
- Not suitable for dense or ill-conditioned matrices.
- Full coefficient extraction negates speedup.

Conclusion

Quantum computing offers a fundamentally new approach to solving linear regression problems by exploiting superposition, entanglement, and quantum parallelism. By reformulating regression as a linear system and leveraging the Harrow–Hassidim–Lloyd (HHL) algorithm, quantum linear regression can theoretically achieve **exponential speedup** over classical methods for large, sparse, and well-conditioned datasets.

This paper discussed the mathematical formulation, quantum state encoding, algorithmic steps, and complexity analysis of quantum linear regression. While practical implementation remains limited by current quantum hardware, QRAM requirements, and challenges in extracting full classical solutions, the approach provides a promising framework for accelerating high-dimensional data analysis and machine learning tasks.

Future research should focus on **hybrid quantum-classical algorithms, error-resilient quantum implementations, and efficient data encoding techniques** to bridge the gap between theoretical speedup and practical applications. Overall, quantum linear regression represents a significant step toward **scalable and efficient machine learning in the era of quantum computing**.

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